SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING **COURSE CODE: CSA43**

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A. <div>
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP C. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I **P Madhavi(192372133)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *P Madhavi*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Question 5: Choose the correct protocol for web communication and justify your choice.

Answer:

Correct Option: B. HTTP (HyperText Transfer Protocol)

Explanation:

HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between a web browser (client) and a web server. It is the foundation of data communication on the World Wide Web (WWW). Whenever a user enters a website address in a browser, clicks on a hyperlink, submits a registration form, watches an online video, or interacts with any web application, the browser sends an **HTTP request** to the web server. The server receives the request, processes it, and returns an **HTTP response** containing the requested resources such as HTML pages, CSS files, JavaScript files, images, videos, or JSON data.

In the given scenario, the startup is developing a modern web application that includes a homepage, registration form, navigation menu, XML configuration files, and server communication. For all these operations, **HTTP** is the most suitable protocol because it is specifically designed for web communication. Every interaction between the client and server follows the HTTP request-response model. For example, when a user opens the homepage, the browser sends an HTTP GET request to retrieve the webpage. When the user submits the registration form, the browser sends an HTTP POST request to the server. The server then validates the data, stores it in the database, and sends an HTTP response indicating whether the registration was successful or not.

Although HTTP is the standard protocol, modern web applications use **HTTPS (HyperText Transfer Protocol Secure)** instead of plain HTTP. HTTPS is an enhanced and secure version of HTTP that encrypts all communication using **SSL (Secure Sockets Layer)** or **TLS (Transport Layer Security)**. This encryption ensures that sensitive user information such as usernames, passwords, email addresses, phone numbers, banking details, and payment information cannot be intercepted or modified by attackers during transmission. HTTPS also verifies the identity of the website through digital certificates, helping users confirm that they are communicating with a legitimate server rather than a fake or malicious website.

Using HTTPS provides several important security benefits. It protects the confidentiality of user data through encryption, ensures data integrity by preventing unauthorized modifications during transmission, and provides authentication through SSL/TLS certificates issued by trusted Certificate Authorities (CAs). Because of these advantages, browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox label HTTP websites as **"Not Secure"** and encourage developers to use HTTPS. Most modern websites, including online banking systems, e-commerce platforms, educational portals, healthcare systems, and social media applications, use HTTPS to safeguard user information.

Why the Other Options Are Incorrect

A. FTP (File Transfer Protocol)

FTP is a protocol designed for transferring files between computers over a network. It is commonly used by developers and administrators to upload website files from a local computer to a web server or to download files from a server. Although FTP is useful for website maintenance, it is **not** used for communication between web browsers and web servers during normal website usage. It cannot display webpages, process forms, or support interactive web applications. Therefore, FTP is not suitable for web communication in this scenario.

C. SMTP (Simple Mail Transfer Protocol)

SMTP is a protocol used specifically for sending emails between email clients and mail servers or between different mail servers. Whenever a user sends an email through services such as Gmail or Outlook, SMTP handles the delivery of the email. However, SMTP cannot be used to retrieve webpages, load website content, or process browser requests. Since the startup's application requires communication between users' browsers and the web server, SMTP is not the appropriate protocol.

D. SNMP (Simple Network Management Protocol)

SNMP is a protocol used for monitoring and managing network devices such as routers, switches, printers, firewalls, and servers. Network administrators use SNMP to collect information about device performance, bandwidth usage, CPU utilization, memory usage, and network status. SNMP plays an important role in network management but has no role in delivering webpages or enabling browser-server communication. Therefore, it is not suitable for web application communication.

Advantages of HTTP/HTTPS

- Enables communication between web browsers and web servers.
- Supports the request-response model used by all websites.
- Transfers web resources such as HTML, CSS, JavaScript, images, videos, and APIs.
- Supports different HTTP methods like GET, POST, PUT, DELETE, and PATCH.
- HTTPS encrypts transmitted data using SSL/TLS.
- Protects user credentials, personal information, and payment details from attackers.
- Provides authentication through digital certificates.
- Ensures data integrity by preventing unauthorized modifications.
- Supported by all modern web browsers and web servers.
- Forms the foundation of almost every web application on the Internet.

Conclusion

The correct answer is **B. HTTP**, as it is the standard protocol used for communication between web browsers and web servers. It enables the transfer of webpages, multimedia content, and user data through the request-response model. In modern web development, **HTTPS** is preferred because it provides encryption, authentication, and data integrity, making web applications secure and reliable. Since the startup's application requires secure communication between users and the server,

HTTP/HTTPS is the most appropriate protocol, while FTP, SMTP, and SNMP are designed for different purposes and cannot replace HTTP in web communication.